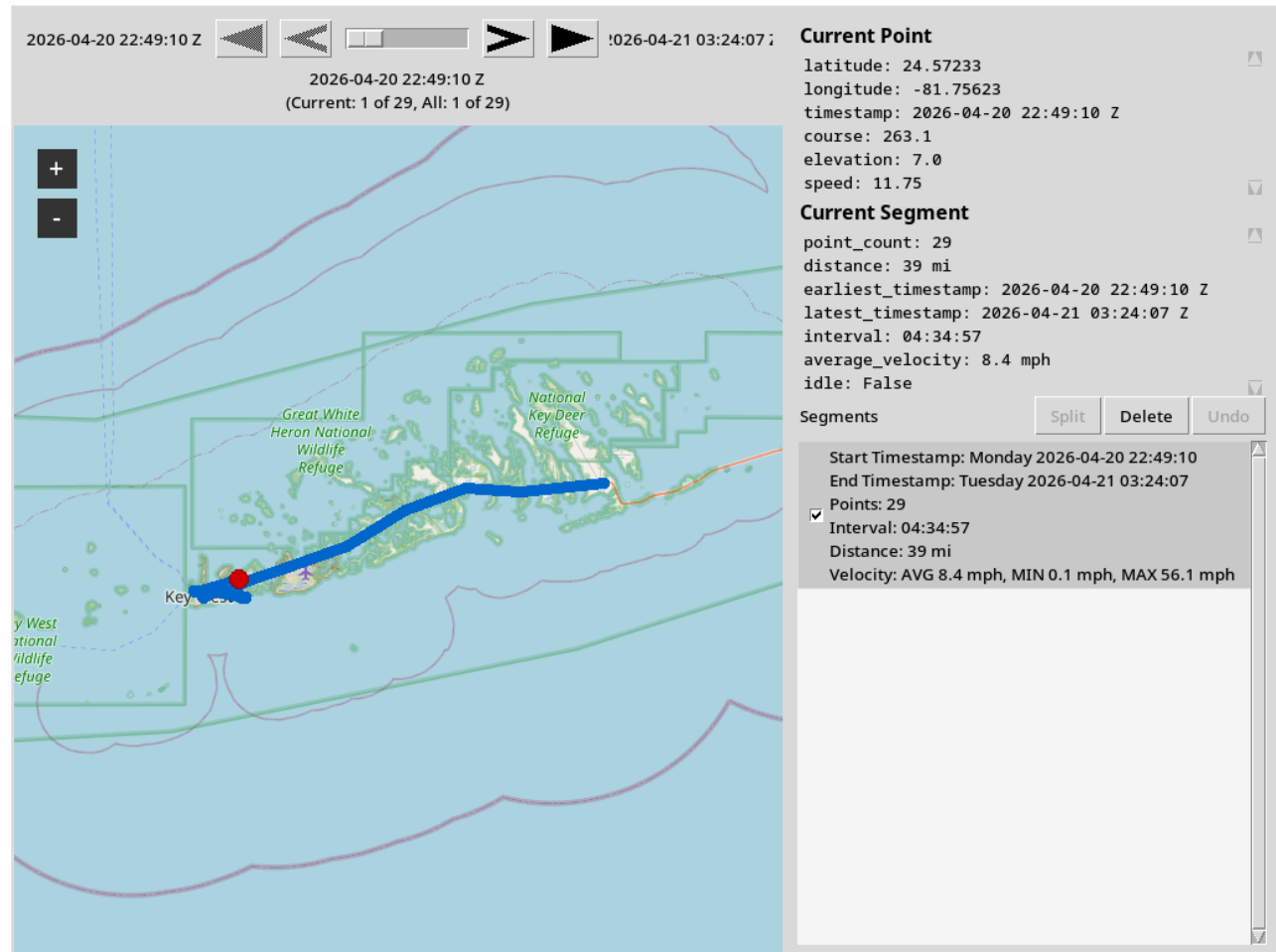


GIS Graphical Editor

Desktop application for viewing GPX tracks on an interactive OpenStreetMap map. Load a GPX file, inspect the path, split or delete segments along the time slider, and optionally overlay recorded points or interval markers along time and distance.



Requirements

- Python 3.10 or newer
- Tkinter (included with most Python installations on Linux)

Installation

From the repository root:

```
python -m venv .venv
source .venv/bin/activate
pip install -e ".[test]"
```

This installs the gge console script and optional pytest dependencies.

Quick start

Launch the application:

```
gge
```

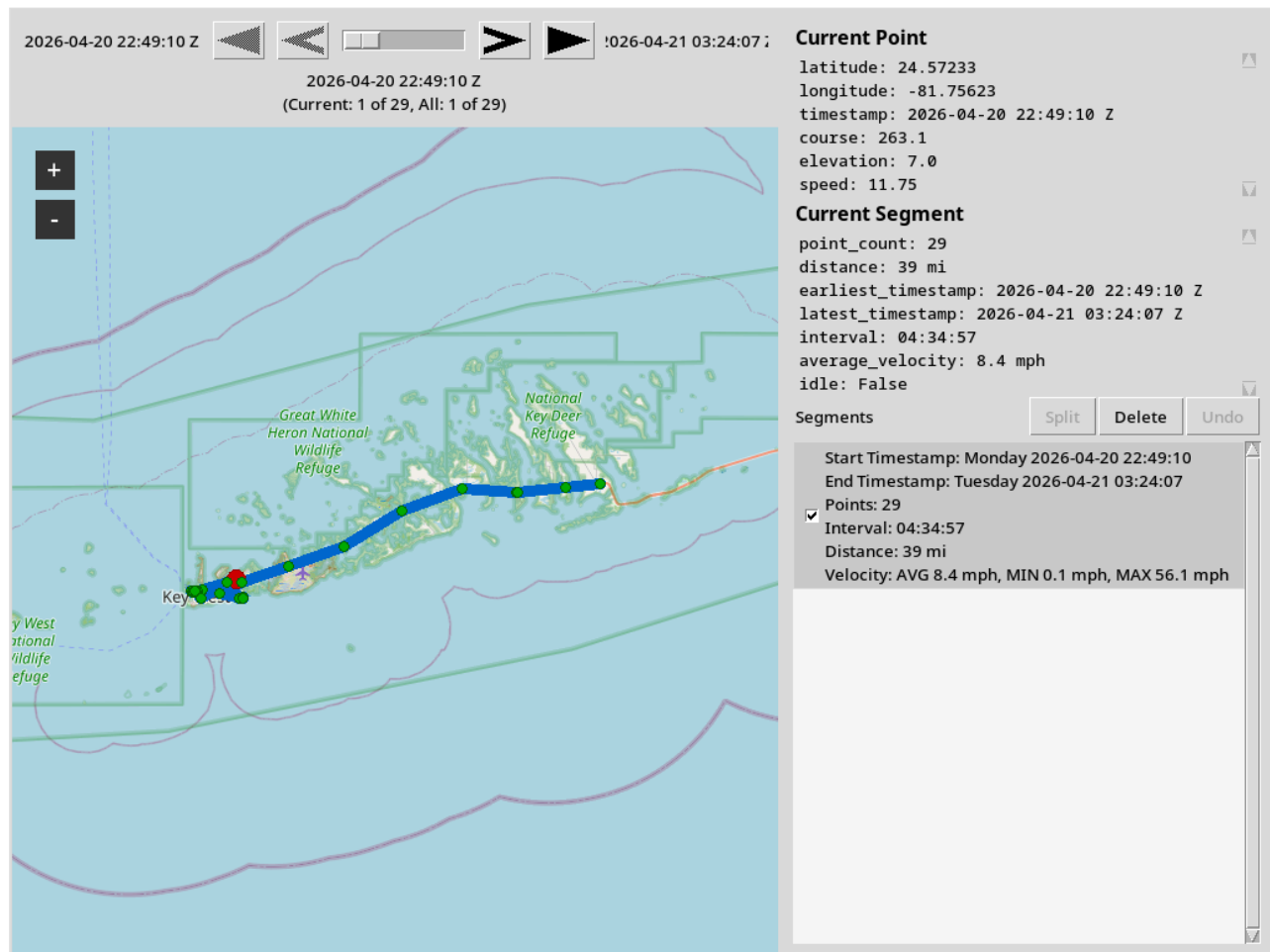
Without `--filepath`, the **File → Load** dialog opens on startup. Choose a `.gpx` file to display the track.

Load a specific file immediately:

```
gge --filepath /path/to/track.gpx
```

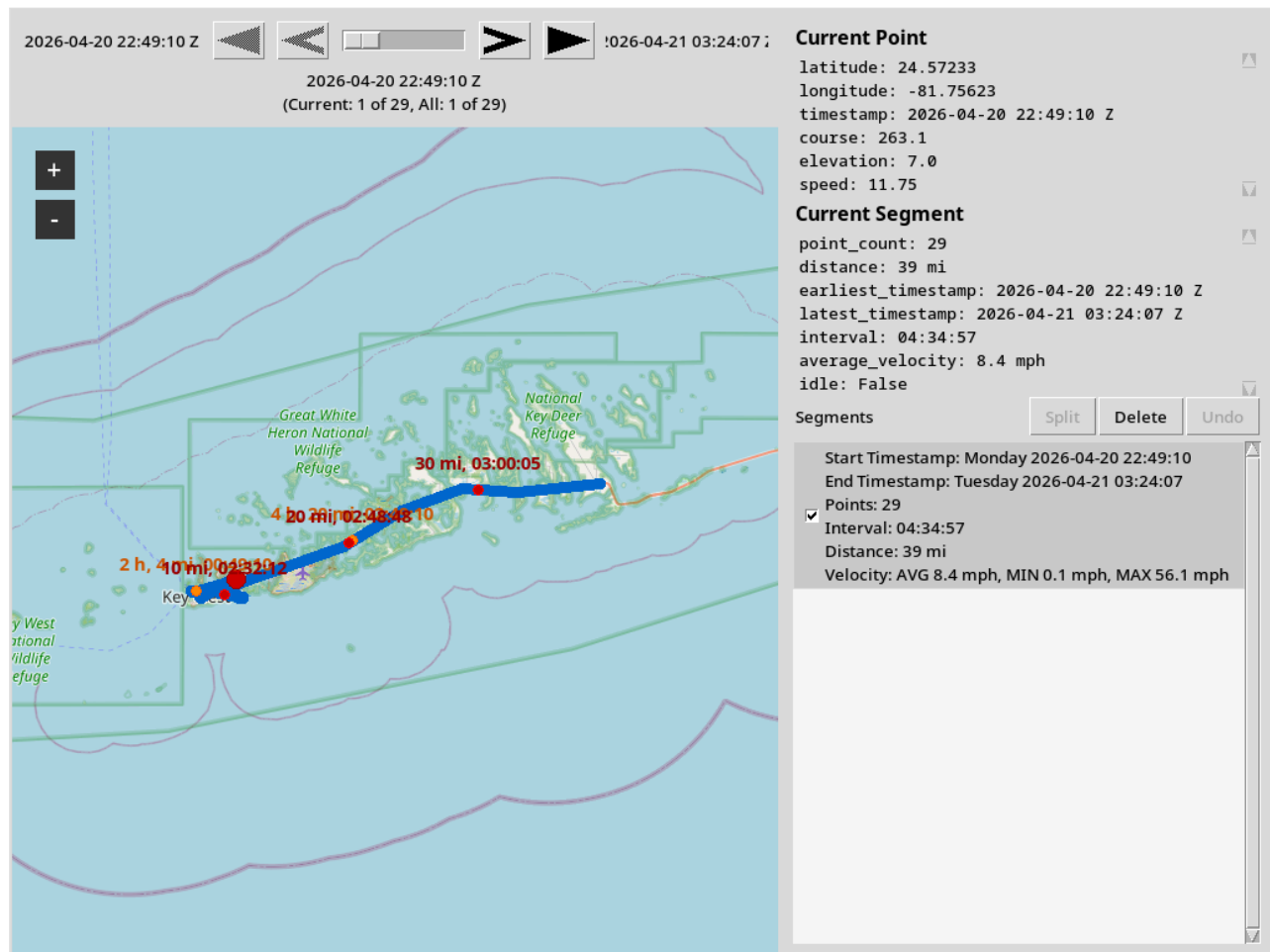
Show every recorded point as a green dot:

```
gge --filepath /path/to/track.gpx --points
```

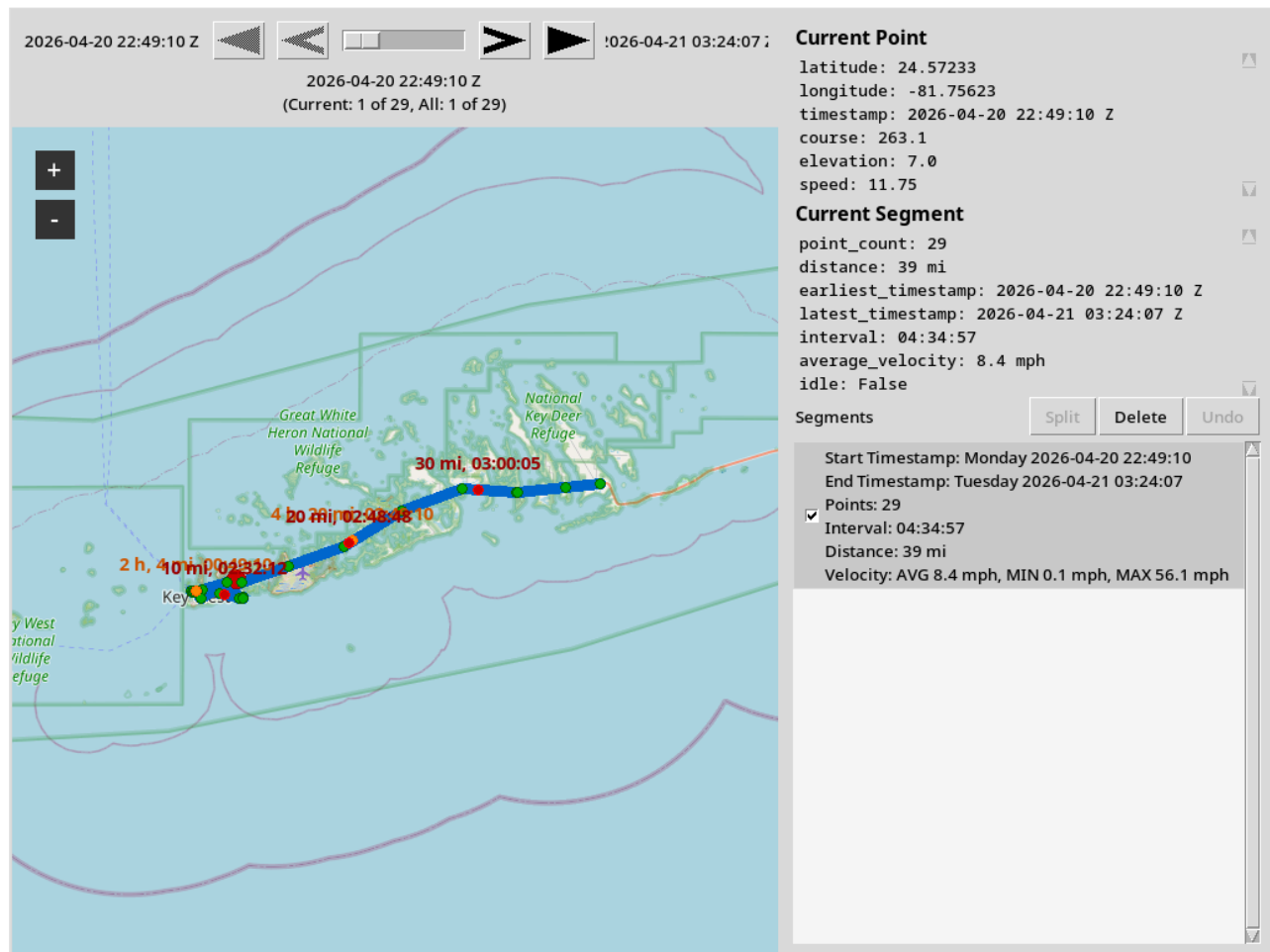


Place orange markers every 2 hours and red markers every 10 miles:

```
gpx --filepath /path/to/track.gpx --mark-hours 2 --mark-distance 10
```



All three overlay types together:

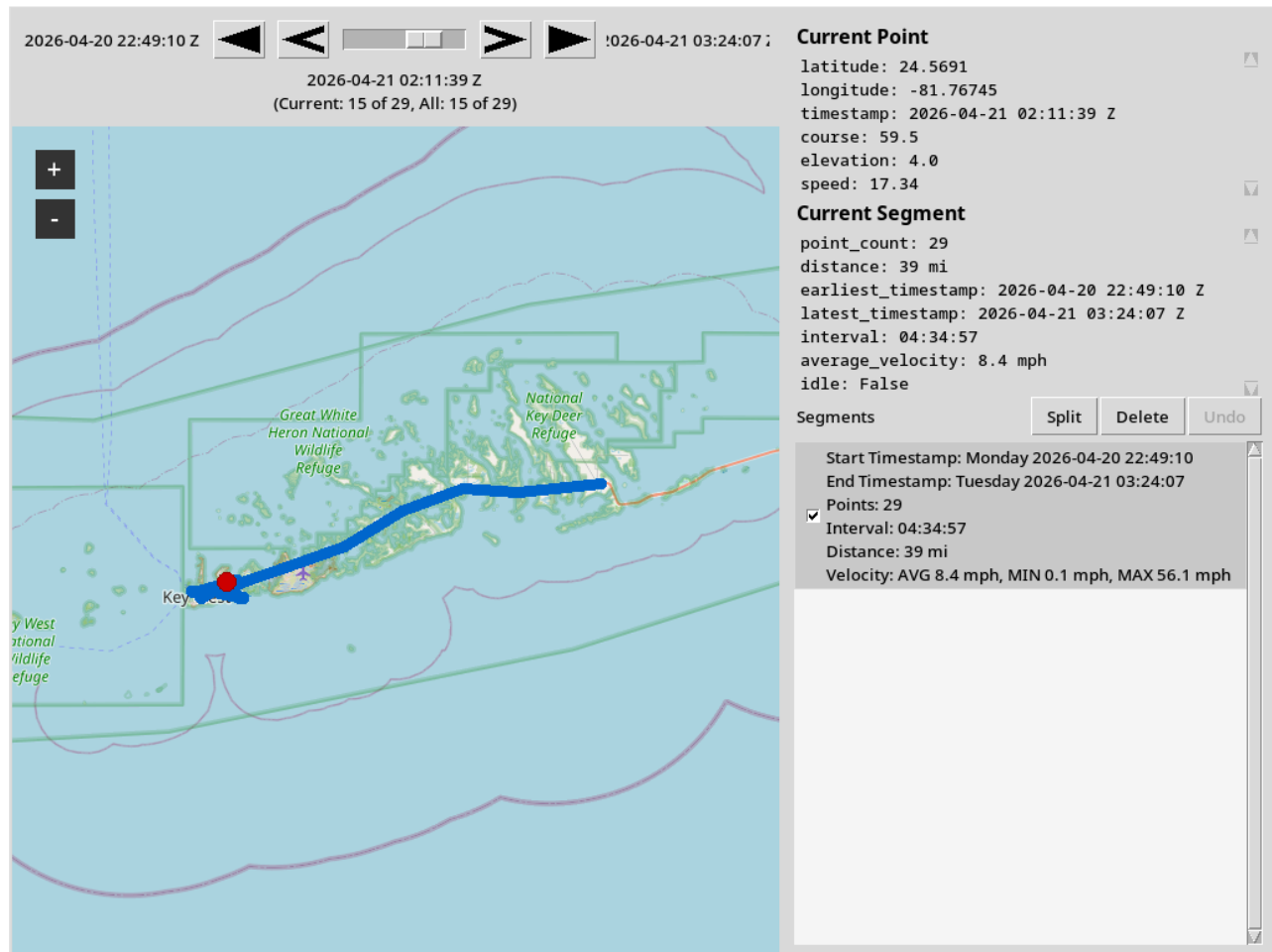


Command-line options

Option	Description
--filepath PATH	Load this GPX file on startup instead of prompting
--points	Draw a green dot at every recorded GPX point
--mark-hours N	Place an orange marker every N hours along the path (requires timestamps)
--mark-distance N	Place a red marker every N miles along the path
--no-mark-labels	Hide text labels on hour and distance interval markers

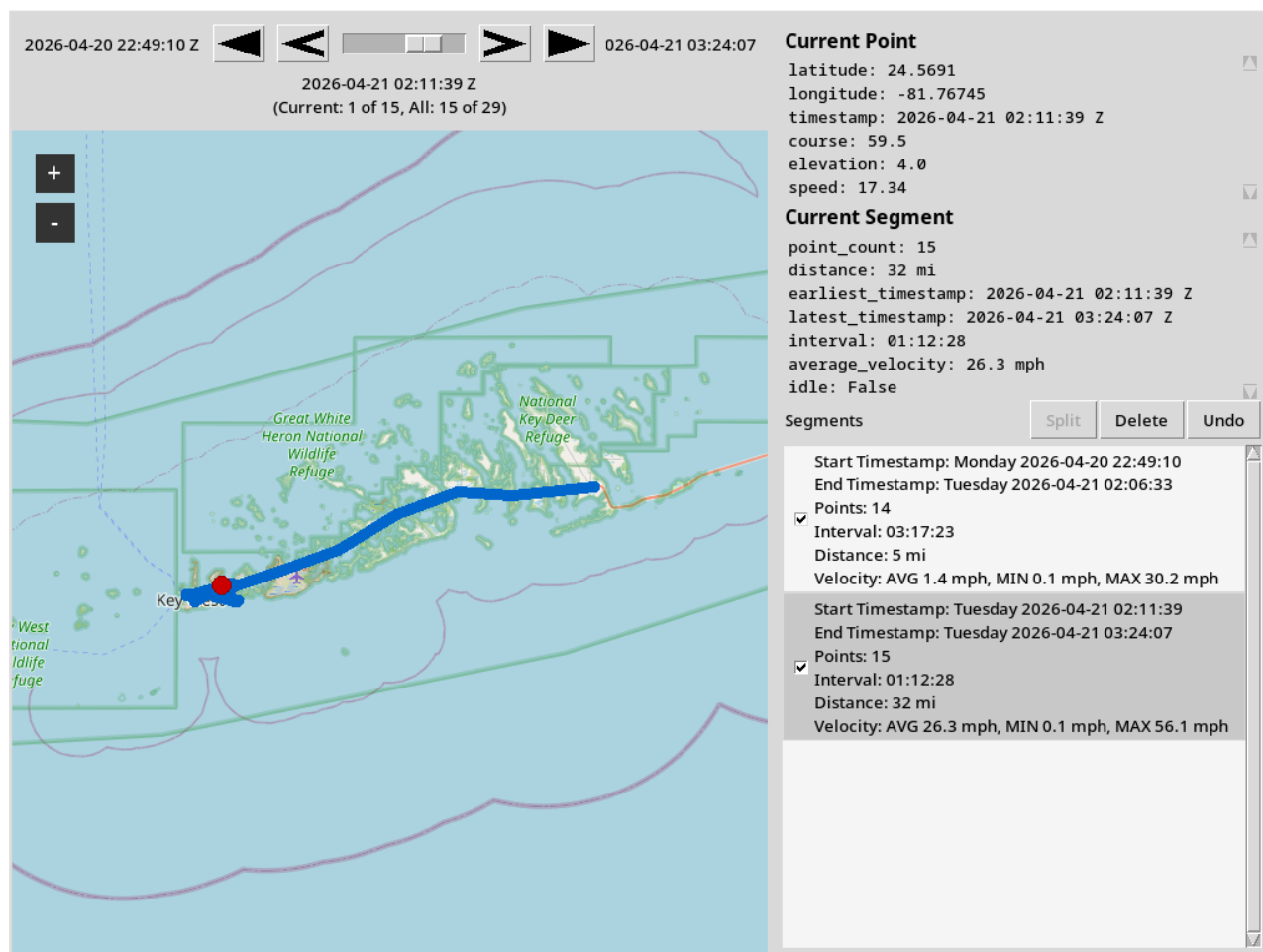
--mark-hours and --mark-distance require positive integers. Run gge --help for full usage text.

Using the map



- **Pan and zoom** using the map widget controls (provided by [tkintermapview](#)).
- **Double-click** the map to zoom in centered on the pointer.
- On load, the view **fits the track bounding box** automatically.
- **Time slider** (when the GPX has timestamps): scrub or use VCR-style controls to animate along the path; a red marker shows the current position.

Track segments



- **Segment checklist** in the right sidebar toggles which segments are drawn on the map.
- **Split** divides the current segment at the slider's nearest point; **Delete** removes the highlighted segment after confirmation.
- **Undo** restores the segment structure from before the last split or delete.
- **Save As** (Ctrl+Shift+S) exports all in-memory segments, including edits.

File menu

Action	Shortcut	Description
Load	Ctrl+O	Open a GPX file via file dialog
Save As	Ctrl+Shift+S	Export the current track (including segment edits) to a GPX file
Close	Ctrl+W	Remove the current map and track (disabled until a file is loaded)
Exit	Ctrl+Q	Quit the application

GPX support

The loader reads points in this order:

1. Track segment points (primary)
2. Route points (when no track segments exist)
3. Standalone waypoints (when tracks and routes are absent)

Coordinates are kept as decimal degrees. Timestamps are used for hour-based markers and distance-marker labels when present.

Development

Run the test suite:

```
pytest -q
```

Project layout

src/gis_graphical_editor/	
entrypoint/gge.py	CLI entry and argument parsing
main_window.py	Tkinter shell, menus, map display
gpx_utility.py	GPX parsing
track_analysis.py	Hour/distance interval markers
track_display_options.py	CLI-driven display settings
map_icon_utility.py	Marker icon images
tests/	Pytest modules mirroring src layout

Feature reference

See [FEATURES.md](#) for a complete list of application capabilities.